Swayam Prakash Yadav

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EDUCATION

Vellore Institute of Technology **Jun 2024 – Jun 2028**
B.Tech CSE (Cyber Security) ***(CGPA: 8.97)*** *Vellore, India*

IIT Madras (Online) **Mar 2024 – Present**
BS Data Science and Applications ***(CGPA: 7.52)***

PROJECTS

Sandboxd | Go, nsjail, Docker, Linux Namespaces, REST API
- Developed a production-oriented sandbox service in Go for securely executing untrusted code with Linux namespaces, cgroups, chroot isolation, and configurable resource limits.
- Designed a declarative runtime architecture where language toolchains, build steps, execution commands, limits, and compiler restrictions are defined entirely through YAML configuration.
- Implemented secure build and execution pipelines with compiler flag allow/deny lists, per-request resource policies, and authenticated REST APIs.
- Built a pluggable sandbox abstraction enabling multiple execution providers while enforcing least-privilege isolation through scoped mounts and minimal Linux capabilities.

Kargo Backend | Kubernetes, Docker, OAuth2, MongoDB, GitHub API
- Backend service for Kargo — a platform enabling containerized app deployments with AI-powered setup and Kubernetes-native scaling.
- Implemented multi-provider authentication (Google OAuth, GitHub OAuth, local login) with secure session management.
- Integrated GitHub APIs for repository linking, deployment automation, and Docker-based build workflows.
- Contributed backend modules for multi-tenant resource allocation, monitoring endpoints, and deployment orchestration.

PlugFit | FastAPI, Celery, Redis, PostgreSQL, Gemini API, MCP
- Built a multi-tenant SaaS platform that acts as a quality layer for MCP servers, ingesting API specs and serving cleaned, improved endpoints.
- Architected a custom executor registry and MCP transport layer to serve cleaned specs as live, tenant-isolated MCP endpoints.
- Designed an LLM-based cleaning pipeline (Gemini 2.0 Flash) with dynamic resource-vocabulary extraction and async job processing via Celery and Redis.
- Implemented a secure auth system with JWT access tokens, httpOnly refresh-token rotation, Google OAuth2, and magic-link verification.

TECHNICAL SKILLS

Languages: Python, Go, JavaScript (ES6), C, C++, SQL, NoSQL
Backend & APIs: FastAPI, Flask, Express.js, Gin (Go), REST APIs, OAuth2, Webhooks
Databases: MySQL, PostgreSQL, MongoDB, SQLite, Redis (Caching, Queues, Messaging)
Infrastructure & DevOps: Docker, Docker Compose, Kubernetes (Basics), GitHub Actions (CI/CD), Nginx, Linux, SSH
Cloud & Deployment: Containerized deployments, GitHub Pages, HuggingFace Spaces, Render
AI & Automation: Google GenAI, Pydantic AI, Prompt Engineering, LLM Workflows, API Automation
Distributed Systems: Asynchronous tasks, Celery, Task queues, Background workers
Developer Tools: VS Code, Postman, Git, Browser DevTools, Make, cURL

CERTIFICATIONS

Introduction to Cybersecurity – Cisco Networking Academy **Jan 2026**
Foundations of Log Analysis for Cyber Defense – Red Team Leaders **Jan 2026**
Networking Basics – Cisco Networking Academy **2026**
AWS Cloud Foundations Training – Amazon Web Services **2026**